Rodrigo Coimbra

Robotics Research Engineer

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Professional Profile

Research Engineer at the Institute for Systems and Robotics, Instituto Superior Técnico, developing autonomous navigation and manipulation capabilities for humanoid robots and mobile manipulators. Research background in reinforcement learning for full-body motion planning and control, with experience integrating robot software for live demonstrations and collaborating with international research teams.

Professional Experience

2025–present **Research Engineer, Institute for Systems and Robotics (ISR)**
Instituto Superior Técnico — SocRob@Home
- Develop autonomous navigation capabilities for a humanoid robot for domestic robotics tasks.
- Contribute to the mobile manipulator's manipulation pipeline and integrate components from partner research teams.
- Collaborate on robotics capabilities and live demonstrations within FOMO-HODOR and euROBIN.

Jul–Sep 2024 **Software Developer Intern, Quidgest**
- Developed a REST API providing asynchronous access to financial-management database data.
- Implemented data processing in C# and SQL queries and updates using Microsoft SQL Server.

Projects and Research

2026–present **euROBIN — Robust pick-and-place demonstration**
- Developed a state machine coordinating perception, motion planning, grasping, and recovery, integrating components from European research partners.
- Added trajectory validation, shortest-valid-plan selection, grasp retries, and recovery from detection and reachability failures.
- Supported four 10-minute demonstrations at the European Parliament in Brussels, with repeated picks and recovery from failures without operator intervention.

2025–present **FOMO-HODOR — Humanoid robotics**
- Implemented RTAB-Map mapping, autonomous navigation, and obstacle avoidance for the LOLA humanoid; contributed basic manipulation capabilities.
- Collaborated with researchers in a project involving ISR, INESC-ID, and the University of Texas at Austin.

2025 **M.Sc. research — Learning-based full-body planning**
- Researched and implemented reinforcement learning for full-body motion planning and control of a mobile manipulator; thesis grade: 18/20.

Technical Skills and Languages

Robotics	ROS, ROS 2, autonomous navigation, motion planning, manipulation, RTAB-Map, Isaac Sim, Isaac Lab.
AI & research	Reinforcement learning, full-body planning, experimental validation.
Programming	Python, C++, C, C#, Java, MATLAB, SQL.
Software tools	Linux, REST APIs, Microsoft SQL Server, LaTeX.
Languages	Portuguese (native); English (C1).

Demonstrations and Events

2026

euROBIN demonstration, European Parliament, Brussels
Took part in a coopera- tive demonstration with other university robotics teams from the euROBIN project, showcasing domestic robotic tasks to members of the European Parliament as an overview of the current state of European robotics.

RoboCup@Home, Incheon, South Korea
Participated as a team member of SocRob in the RoboCup@Home competition, an international event that brings together hundreds of re- searchers and teams working on service and domestic robotics.

ICARSC, Barcelos, Portugal
Attended the International Conference on Autonomous Robot Systems and Competitions, where I presented my research on full-body local planning for a mobile manipulator.

Education

2023–2025

M.Sc. in Electrical and Computer Engineering
Instituto Superior Técnico — Final grade: 18/20.
Specialisation in control, robotics, and artificial intelligence; secondary focus on computing systems. During the final year of my master's degree, I joined the SocRob@Home team, where I deepened my knowledge and practical experience in robotics. Academic Merit Certificate in both academic years.

2020–2023

B.Sc. in Electrical and Computer Engineering
Instituto Superior Técnico — Final grade: 16/20.
Academic Merit Certificate in all three academic years.

Publication

1 **R. Coimbra**, R. Bettencourt, A. Vale, and P. U. Lima, "Full-body local planning of a mobile manipulator using reinforcement learning," in *2026 IEEE International Conference on Autonomous Robot Systems and Competitions (ICARSC)*, IEEE, 2026, pp. 9–14. URL: https://ieeexplore.ieee.org/document/11523319

Science Communication

2024–2026

Robotics demonstrations and research outreach, ISR
Presented robotics research and ran live demonstrations for visiting schools and the public. Activities included JEEC, Técnico Days, European Researchers' Night, and events at Pavilhão do Conhecimento.